

Kinematic Wave Theory

Zhengbing He, Ph.D.

<https://www.GoTrafficGo.com>

January 2, 2026

- Introduction
- Conservation law
- Initial and boundary conditions
- Homogeneous problems
- Summary

- **Introduction**
- Conservation law
- Initial and boundary conditions
- Homogeneous problems
 - A special case
 - General cases
- Summary

Traffic flow theory

- Dynamic macroscopic models (**Continuum models**)
- Kinematic wave theory (**LWR model**)

- **Continuum models** assume that Δx and Δt can be replaced by **differentials** (i.e., dx and dt - treating finite intervals as infinitesimally small differentials), which open the door to the world of **calculus**; i.e. assumption that a relationship between flow and density existed at all locations and times. The traffic is viewed as a **fluid** with a well-defined density everywhere;



Traffic flow theory

- Dynamic macroscopic models (**Continuum models**)
- Kinematic wave theory (**LWR model**)

- **Continuum models** assume that Δx and Δt can be replaced by **differentials** (i.e., dx and dt - treating finite intervals as infinitesimally small differentials), which open the door to the world of **calculus**; i.e. assumption that a relationship between flow and density existed at all locations and times. The traffic is viewed as a **fluid** with a well-defined density everywhere;
- **KWT** (Lighthill and Whitham, 1955 and Richards, 1956; thus also LWR) is the simplest one of the continuum models ($\therefore$ first-order).
- The **goal** is to **predict** the **evolution** of traffic from **initial conditions**;

A Description of LWR

- Lighthill and Whitham (1955) and Richards (1956) provided the first **traffic flow approximation models** that compared traffic flow to **fluid flow**. LWR is extensively used as the preferred model for representing flow dynamics from a **macroscopic perspective**.
- **CAPABLE** in providing a coarse description of main traffic features (e.g., **formation and dissolution of shockwaves**); **INADEQUATE** in describing complex (i.e., higher-order) traffic patterns **such as stop-and-go waves, capacity drop phenomena**.
- Based on a *hyperbolic partial differential equation* of first order, which describes the conservation of cars in time and space.
- Assumes that the relation q - k - v observed under steady states **holds at all times**, even when flow and density vary with time and space. This assumption **suppresses all other traffic states** and phase transitions not belonging on this curve.

Hyperbolic Partial Differential Equation



Hyperbolic partial differential equation

In mathematics, a **hyperbolic partial differential equation** of order n is a partial differential equation (PDE) that, roughly speaking, has a well-posed initial value problem for the first $n - 1$ derivatives. More precisely, the Cauchy problem can be locally solved for arbitrary initial data along any non-characteristic hypersurface. Many of the equations of mechanics are hyperbolic, and so the study of hyperbolic equations is of substantial contemporary interest. The model hyperbolic equation is the wave equation. In one spatial dimension, this is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

The equation has the property that, if u and its first time derivative are arbitrarily specified initial data on the line $t = 0$ (with sufficient smoothness properties), then there exists a solution for all time t .

The solutions of hyperbolic equations are "wave-like". If a disturbance is made in the initial data of a hyperbolic differential equation, then not every point of space feels the disturbance at once. Relative to a fixed time coordinate, disturbances have a finite propagation speed. They travel along the characteristics of the equation. This feature qualitatively distinguishes hyperbolic equations from elliptic partial differential equations and parabolic partial differential equations. A perturbation of the initial (or boundary) data of an elliptic or parabolic equation is felt at once by essentially all points in the domain.

Although the definition of hyperbolicity is fundamentally a qualitative one, there are precise criteria that depend on the particular kind of differential equation under consideration. There is a well-developed theory for linear differential operators, due to Lars Gårding, in the context of microlocal analysis. Nonlinear differential equations are hyperbolic if their linearizations are hyperbolic in the sense of Gårding. There is a somewhat different theory for first order systems of equations coming from systems of conservation laws.



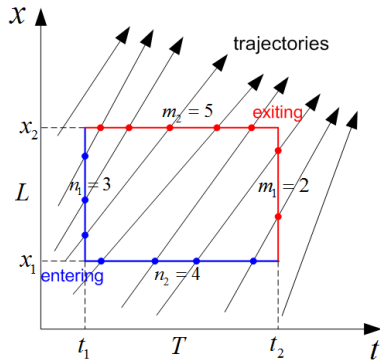
A hyperbola is an open curve with ⁶³ two branches, the intersection of a plane with both halves of a double cone. The plane does not have to be parallel to the axis of the cone; the hyperbola will be symmetrical in any case.

The basic components of the KWT are as follows:

- ① Fundamental diagrams ✓;
- ② Conservation law;
- ③ Initial and boundary conditions.

- Introduction
- **Conservation law**
- Initial and boundary conditions
- Homogeneous problems
- Summary

Conservation Law w/o Lane-Changing



exiting *entering*

$$m_1 + m_2 = n_1 + n_2$$

$$\Rightarrow (m_1 - n_1) + (m_2 - n_2) = 0$$

$$\Rightarrow \frac{m_1 - n_1}{LT} + \frac{m_2 - n_2}{LT} = 0$$

$$\Rightarrow \frac{k_{t_2} - k_{t_1}}{T} + \frac{q_{x_2} - q_{x_1}}{L} = 0$$

$k = \text{veh/km}, q = \text{veh/h}$

$$\Rightarrow \frac{\Delta k}{T} + \frac{\Delta q}{L} = 0 \quad (\Delta : \text{small } L \& T)$$

$$\Rightarrow \frac{\partial k(t,x)}{\partial t} + \frac{\partial q(t,x)}{\partial x} = 0 \quad (\Delta \rightarrow 0)$$

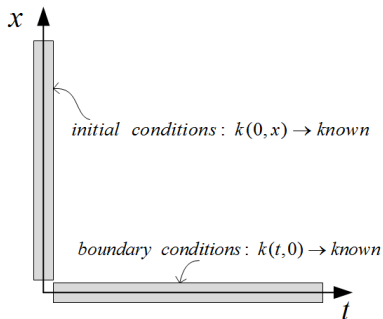
calculus

Hyperbolic PDE

treating finite intervals as
infinitesimally small differentials

Initial and Boundary Conditions

- **Initial conditions:** $k(0, x)$ is known;
- **Boundary conditions:** $k(t, 0)$ is known;



- Introduction
- Conservation law
- Initial and boundary conditions
- **Homogeneous problems**
- Summary

Homogeneous Problem

Assume

(1) homogeneous vehicles (2) no on- or off-ramp (3) Concave FD.

Have

(i) fundamental diagram (ii) conservation law (iii) initial conditions:

$$\left\{ \begin{array}{l} \text{(i) } q = Q(k) \\ \text{(ii) } \frac{\partial k(x,t)}{\partial t} + \frac{\partial q(x,t)}{\partial x} = 0 \\ \text{(iii) } k(x,0) = g(x) \end{array} \right. \longrightarrow \left\{ \begin{array}{l} \frac{\partial k}{\partial t} + Q'(k) \frac{\partial k}{\partial x} = 0 \\ k(x,0) = g(x) \end{array} \right.$$

GOAL: to get $q(x, t)$, $k(x, t)$. Let's solve the system of equations

Homogeneous Problem



WIKIPEDIA
The Free Encyclopedia

Chain rule

In calculus, the **chain rule** is a formula that expresses the derivative of the composition of two differentiable functions f and g in terms of the derivatives of f and g . More precisely, if $h = f \circ g$ is the function such that $h(x) = f(g(x))$ for every x , then the chain rule is, in Lagrange's notation,

$$h'(x) = f'(g(x))g'(x).$$

or, equivalently,

$$h' = (f \circ g)' = (f' \circ g) \cdot g'.$$

The chain rule may also be expressed in Leibniz's notation. If a variable z depends on the variable y , which itself depends on the variable x (that is, y and z are dependent variables), then z depends on x as well, via the intermediate variable y . In this case, the chain rule is expressed as

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx},$$

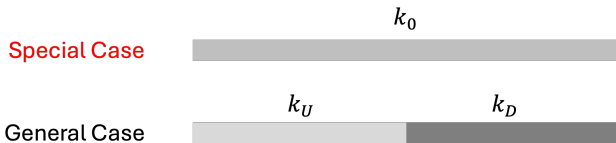
and

$$\left. \frac{dz}{dx} \right|_x = \left. \frac{dz}{dy} \right|_{y(x)} \cdot \left. \frac{dy}{dx} \right|_x,$$

for indicating at which points the derivatives have to be evaluated.

In integration, the counterpart to the chain rule is the substitution rule.

Homogeneous Problem

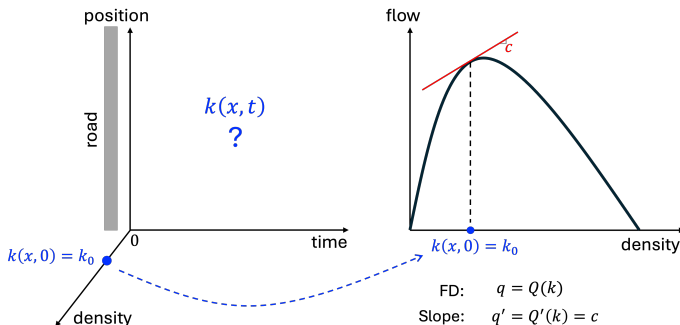


Special Case: Advection Equation

Special case: traffic on a road is in one state, i.e., $k(x, 0) = k_0$

$$Q'(k) = c$$

where $Q'(k)$ is the slope of FD and c is a constant,

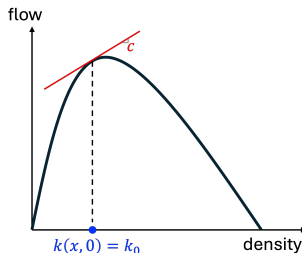
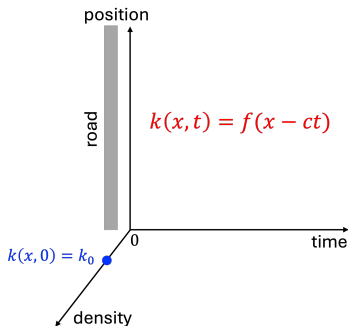


Special Case: Advection Equation

Solve $\frac{\partial k}{\partial t} + c \frac{\partial k}{\partial x} = 0$ **for** k , given (x, t, c) - space, time, slope

$$k(t, x) = f(x - ct)$$

where $f(\cdot)$ is an arbitrary function.



FD: $q = Q(k)$

Slope: $q' = Q'(k) = c$

Special Case: Advection Equation

Verification (chain rule):

$$\begin{cases} \frac{\partial k}{\partial t} = \frac{df(y)}{dy} \cdot \frac{dy}{dt} = \frac{df(y) \cdot (-c)}{dy} \\ c \frac{\partial k}{\partial x} = c \frac{df(y)}{dy} \cdot \frac{dy}{dx} = c \frac{df(y)}{dy} \end{cases} \longrightarrow \frac{\partial k}{\partial t} + c \frac{\partial k}{\partial x} = 0$$

$$y = x - ct$$

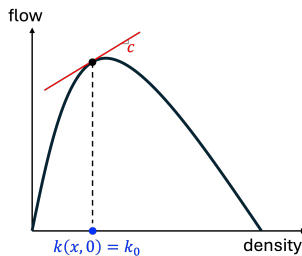
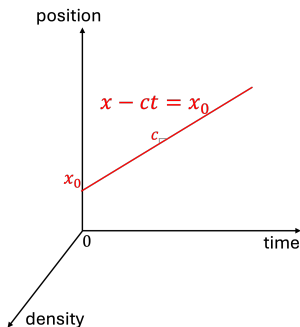
Special Case: Advection Equation

The solution, i.e., $k(x, t) = f(x - ct)$, means that

if $(x - ct)$ is constant, e.g.,

$$x - ct = x_0$$

We can find a straight line on the t-x plane as follows



FD: $q = Q(k)$

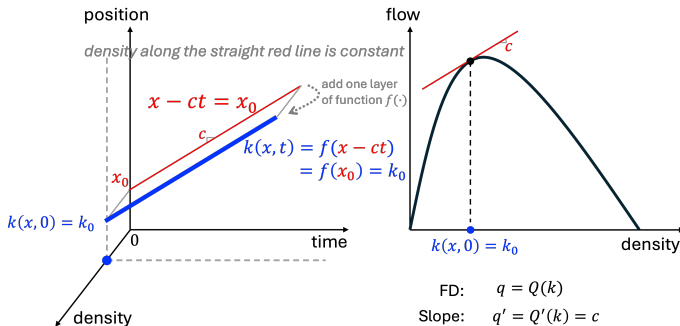
Slope: $q' = Q'(k) = c$

Special Case: Advection Equation

The corresponding density $k(x, t)$ at location x on a road at time t is constant:

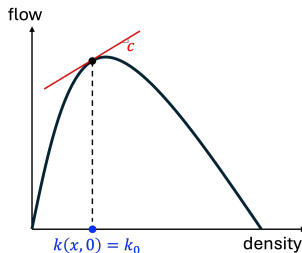
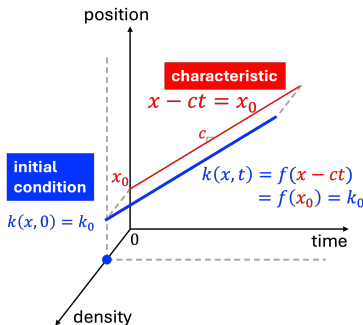
$$k(x, t) = f(\mathbf{x} - \mathbf{c}t) = f(\mathbf{x}_0) = \text{constant} = k_0$$

(Add one more layer of function)



Special Case: Advection Equation

The straight line (red) is called “**characteristic**”: Given initial condition $k(x, 0) = k_0$, we have a characteristic with a slope $c = Q'(k(x, 0))$. Along this characteristic, the density is the same and equal to $k(x, 0) = k_0$.



FD: $q = Q(k)$
Slope: $q' = Q'(k) = c$

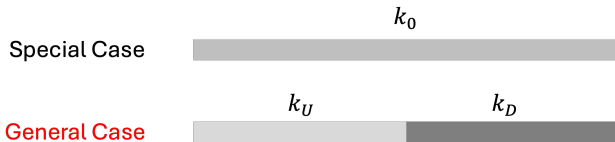
Characteristics (called in mathematics books) are also called “**waves**” or “signals” in the fluids literature.

The properties of characteristics are:

- Density, flow and speed are **CONSTANT** and equal to the values at the boundary;
- Vehicle speed is always **GREATER** than the characteristic speed c ;
- Relations with the fundamental diagram:

traffic states	k	c	direction of characteristics
free flow	$k < k_c$	$c = Q'(k) > 0$	with traffic stream
capacity	$k = k_c$	$c = Q'(k) = 0$	stationary
congestion	$k > k_c$	$c = Q'(k) < 0$	against traffic stream

Homogeneous Problem

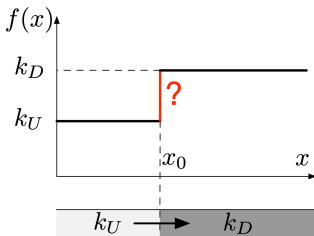


General Cases: Riemann Problem

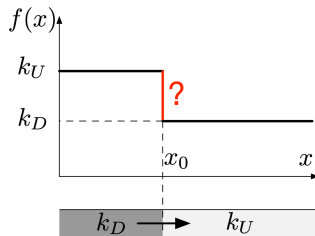
The initial condition is

$$Q'(k) = c(k) = \begin{cases} c_U, & k = k_U, \ x < x_0 \\ c_D, & k = k_D, \ x > x_0 \end{cases}$$

The goal is **how the two traffic states transit:**

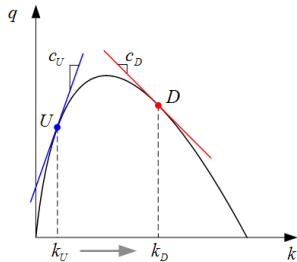
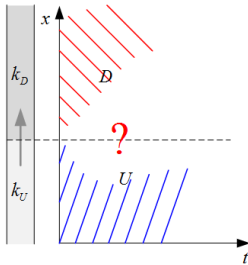


Case I

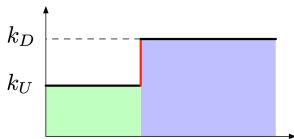


Case II

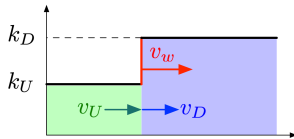
Case I: Shock waves



Case I: Shock waves



Case I: Shock waves



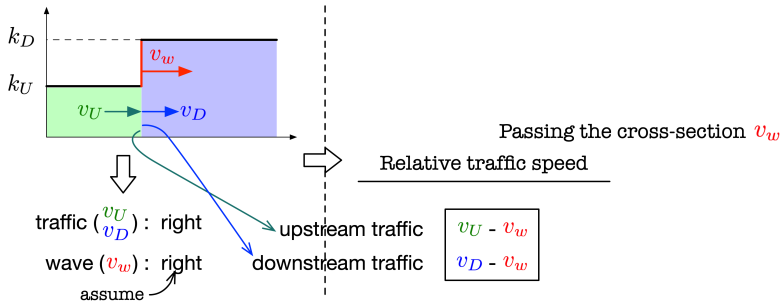
traffic (v_U): right
 (v_D)

wave (v_w): right

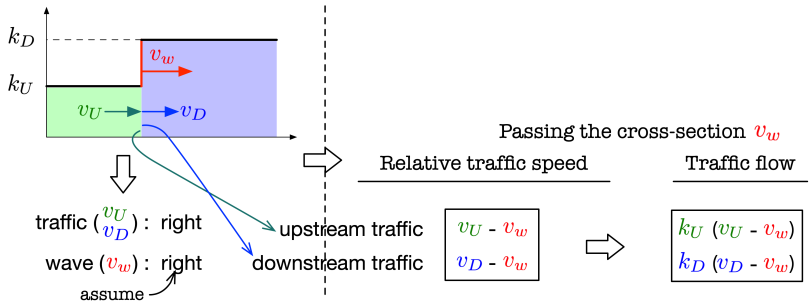
assume



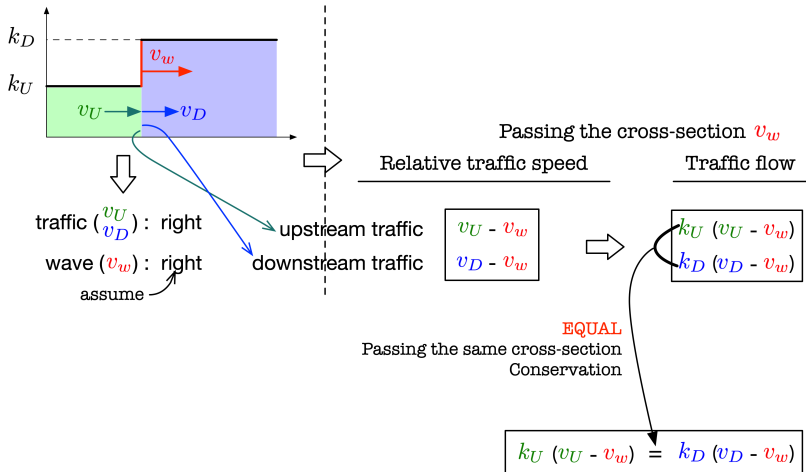
Case I: Shock waves



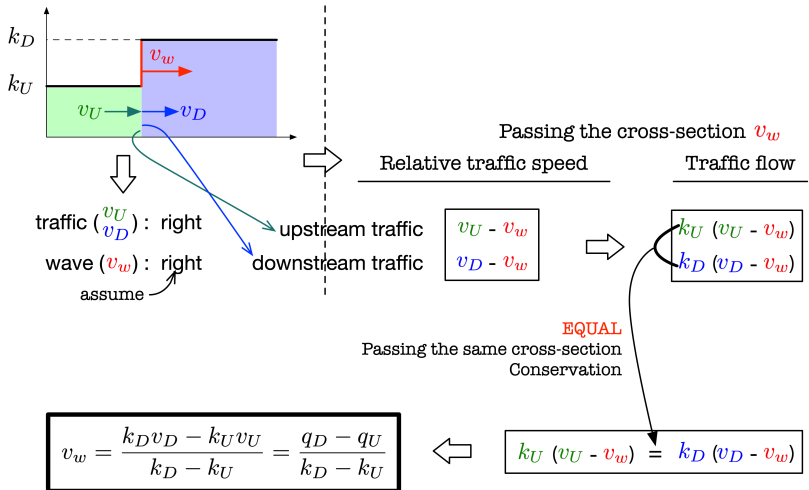
Case I: Shock waves



Case I: Shock waves



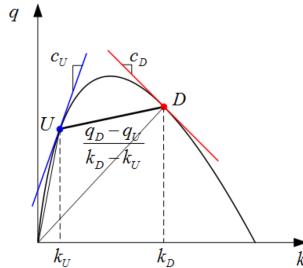
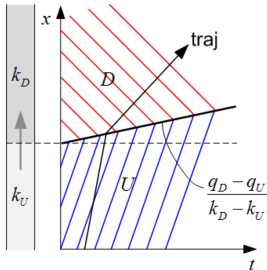
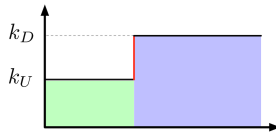
Case I: Shock waves



Case I: Shock waves

Case I: Shock Wave

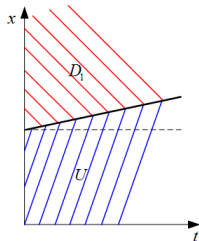
$$k_U < k_D$$



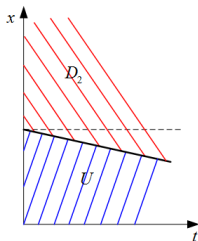
Case I: Shock waves

- Shock waves separate the waves from **lower density** to **higher density**
- Characteristics **end in the shock wave** and the traffic state shows a **discontinuous changes**
- Trajectories that cross a shock wave **change their speed abruptly**

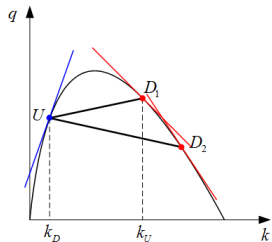
Directions of shock waves



Moving with the traffic

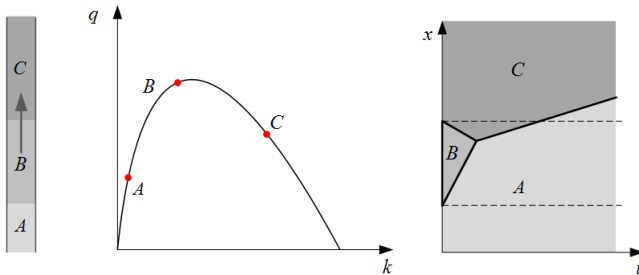


Moving against the traffic



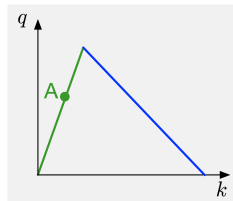
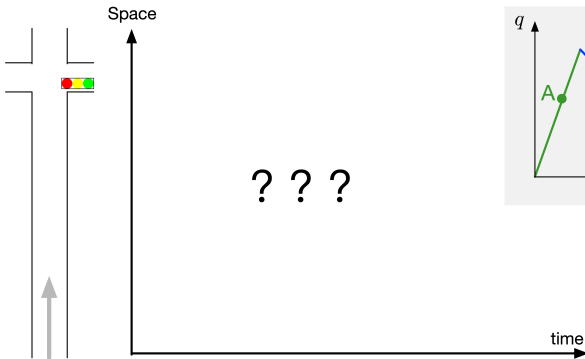
Case I: Shock waves

Practice-1: merging shock waves



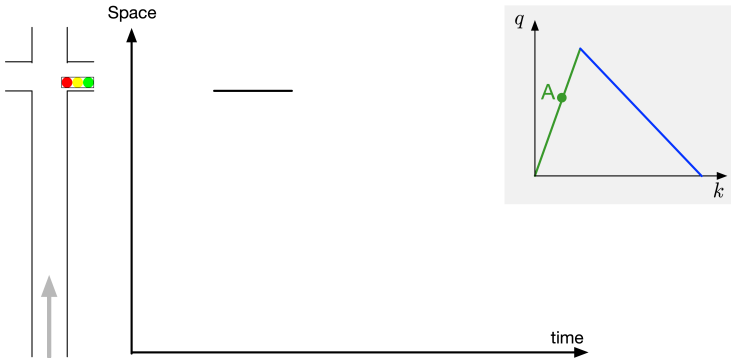
Case I: Shock waves

Practice-2: approaching an intersection



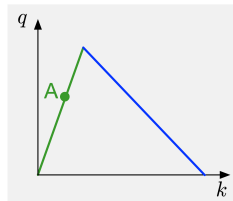
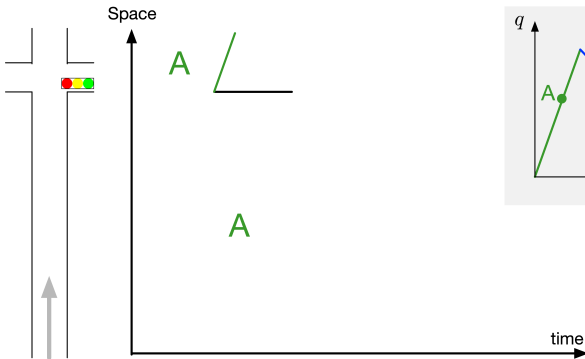
Case I: Shock waves

Practice-2: approaching an intersection



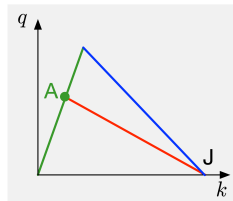
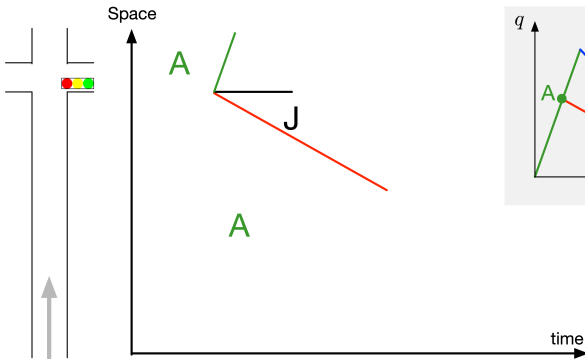
Case I: Shock waves

Practice-2: approaching an intersection



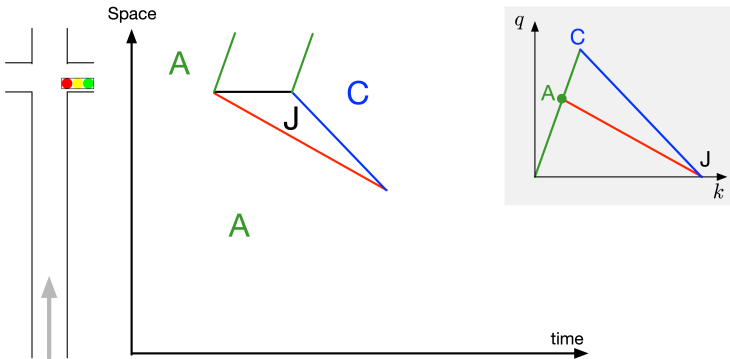
Case I: Shock waves

Practice-2: approaching an intersection



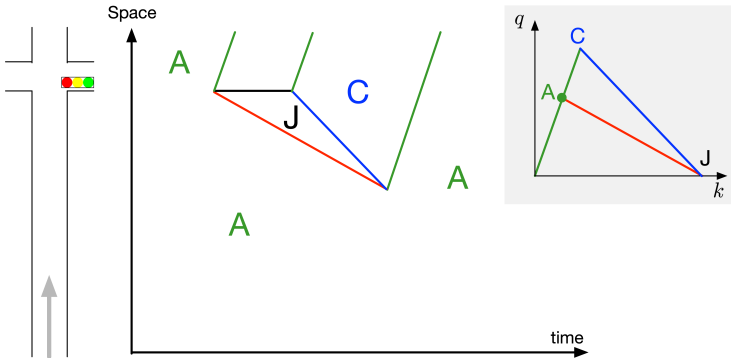
Case I: Shock waves

Practice-2: approaching an intersection



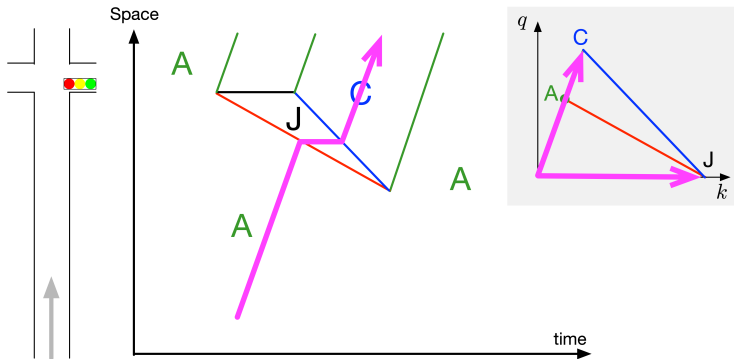
Case I: Shock waves

Practice-2: approaching an intersection

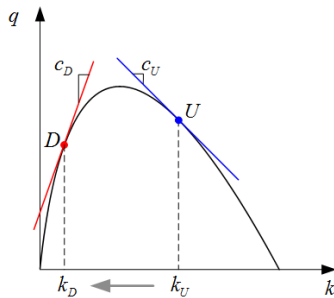
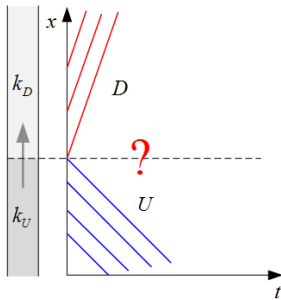


Case I: Shock waves

Practice-2: approaching an intersection

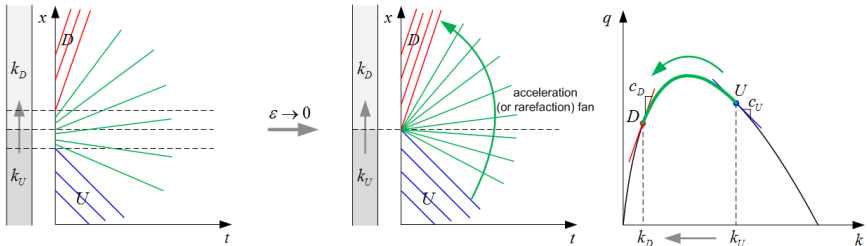
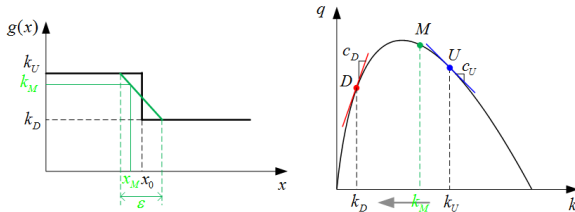


Case II: Fans



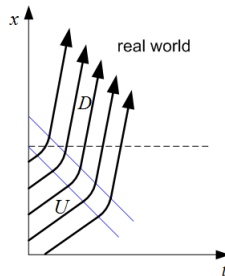
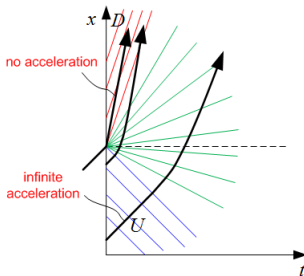
Case II: Fans

(1) expand the process - instantaneous event; (2) contract it back



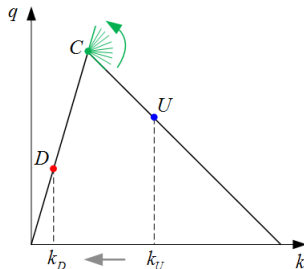
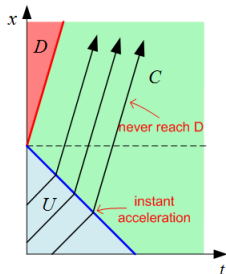
Case II: Weakness of Fundamental Diagrams

- Acceleration does not exist when $t = 0$
- Time to accelerate from v_U to v_D increases with the vehicle number

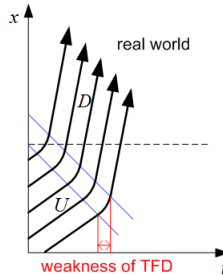
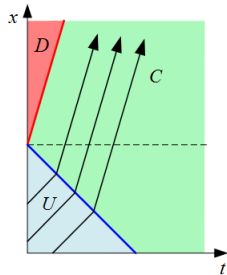


Case II: Triangular Fundamental Diagrams

Traffic state C carries all intermediate characteristics $v_{char} \in [-w, v_f]$



Case II: Weakness of TFDs



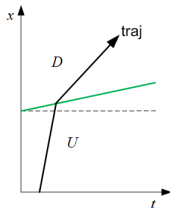
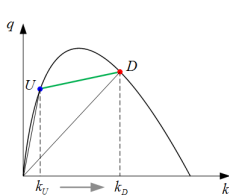
- Introduction
- Conservation law
- Initial and boundary conditions
- Homogeneous problems
- **Summary**

Summary

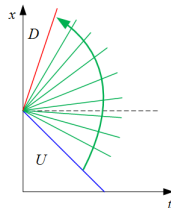
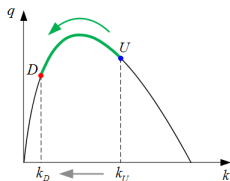
Initial Value Problem (IVP) :
$$\begin{cases} \partial_t k + Q'(k) \partial_x k = 0 \\ k(0, x) = g(x) \end{cases}$$

- **ONE** traffic state: $Q'(k) = c \rightarrow$ Solution: $k(t, x) = f(x - tc)$
- **TWO** traffic states: decompose into a series of Riemann problems

Case I

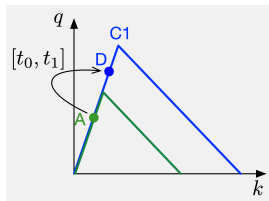
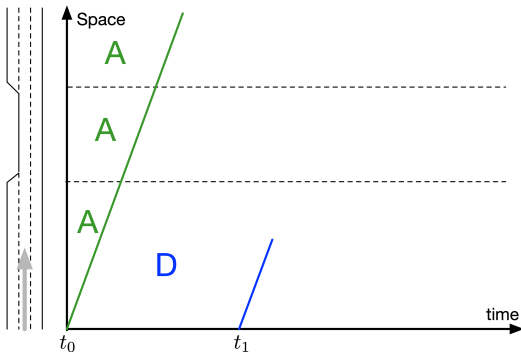


Case II



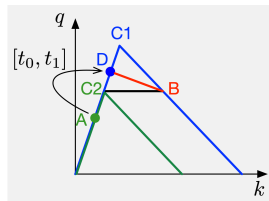
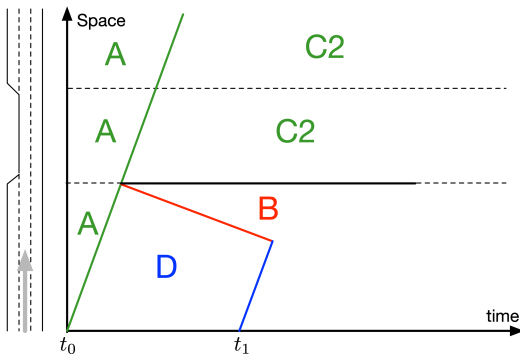
Practice-3

Facing a bottleneck (3 lanes $\rightarrow$ 2 lanes), traffic demand increases to **D** from **A** during t_0 and t_1 . How will the traffic evolve?



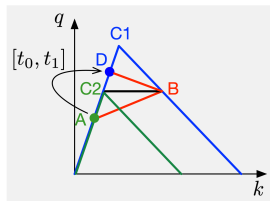
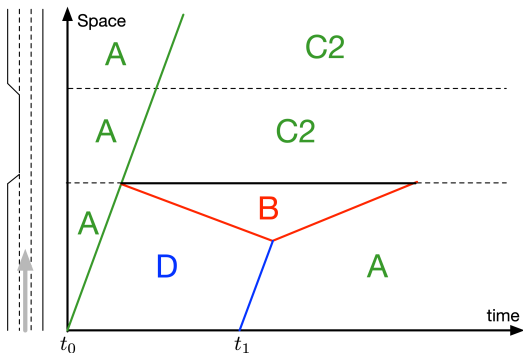
Practice-3

Facing a bottleneck (3 lanes $\rightarrow$ 2 lanes), traffic demand increases to **D** from **A** during t_0 and t_1 . How will the traffic evolve?



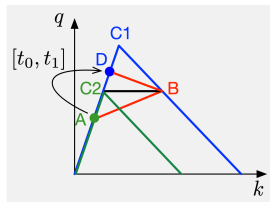
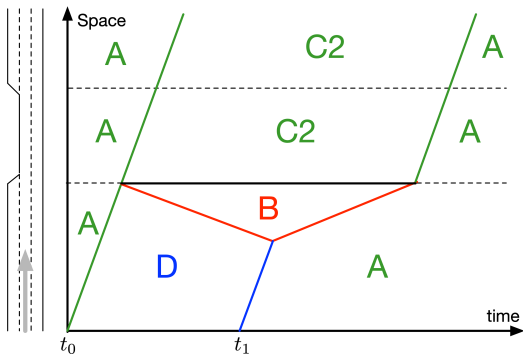
Practice-3

Facing a bottleneck (3 lanes $\rightarrow$ 2 lanes), traffic demand increases to **D** from **A** during t_0 and t_1 . How will the traffic evolve?



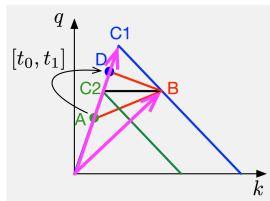
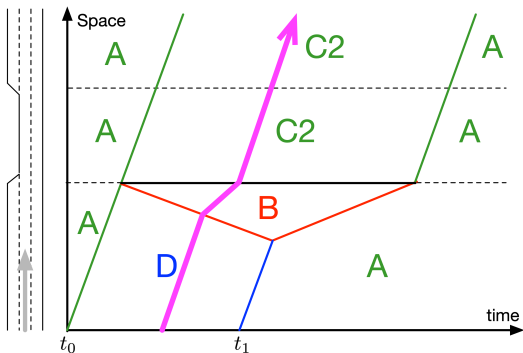
Practice-3

Facing a bottleneck (3 lanes $\rightarrow$ 2 lanes), traffic demand increases to **D** from **A** during t_0 and t_1 . How will the traffic evolve?



Practice-3

Facing a bottleneck (3 lanes $\rightarrow$ 2 lanes), traffic demand increases to **D** from **A** during t_0 and t_1 . How will the traffic evolve?



Thank you!